Literature Review

The following literature review supplies the scientific foundation and the operational baselines that must be met/exceeded. How to handle imbalance and interpretability precedents

Rapid Intensification

Kaplan & DeMaria (2003)

Large-Scale Characteristics of Rapidly Intensifying Tropical Cyclones in the North Atlantic Basin

John Kaplan and Mark DeMaria. 2003. Weather and Forecasting, 18, 1093–1108

The foundational observational study that defined the modern 30-kt / 24-h rapid-intensification threshold for the Atlantic and produced the first simple multi-predictor Rapid Intensification Index later used operationally by the National Hurricane Centre .

Overview

Using the NHC best-track and SHIPS databases for 1989–2000 Atlantic tropical cyclones, the authors defined rapid intensification (RI) as a 24-h over-water intensity increase of ≥30 kt (approximately the 95th percentile).

They showed that:

- Virtually all major and category 4–5 hurricanes experience at least one RI episode
- RI cases differ systematically from non-RI cases in:
    - Location
    - Previous intensity change
    - Proximity to maximum potential intensity
    - Sea-surface temperature

- Humidity
- Vertical shear
- Upper-level flow

A simple threshold-counting technique combining the five most discriminating predictors raised the estimated RI probability from 1% (no thresholds met) to 41% (all five met). The technique was tested in real time during the 2001 season as part of the Joint Hurricane Testbed.

Problem & Motivation

Operational intensity forecasts, especially the prediction of sudden, large intensity increases, remained far less skilful than track forecasts. High-profile near-landfall RI events such as Hurricanes Opal (1995) and Bret (1999) underscored the problem's practical urgency.

Although many case studies examined oceanic, inner-core, and environmental contributions to intensity change, few systematically quantified the large-scale conditions that favour RI across a large sample of Atlantic storms.

The authors therefore set out to establish a statistically robust definition of RI and identify the environmental differences between RI and non-RI cases that could be used as a practical forecasting tool.

Key Concepts & Terminology

- **Rapid Intensification (RI)**

  Defined here as a maximum sustained surface wind increase of ≥15.4 m s^{-1} (30 kt) in 24 h while the storm remains over water. This threshold corresponds to roughly the 95th percentile of all 24-h over-water intensity changes in the 1989–2000 Atlantic sample.

  The difference between a storm that strengthens gradually and one that jumps roughly one Saffir–Simpson category in a single day.

- **Maximum Potential Intensity (MPI) – current intensity (POT)**

  How far the storm is from the theoretical upper limit set by the underlying sea-surface temperature. Larger values indicate greater thermodynamic “room” to intensify.

- **Vertical shear (SHR)**

  Magnitude of the vector difference between the 850- and 200-hPa winds, area-averaged around the storm. Low shear is repeatedly shown to favour RI.

- **Simple threshold technique**

  A transparent, non-regression method that counts how many of the five most discriminating predictors exceed empirically chosen thresholds and converts that count into an RI probability.

Methodology

Dataset

- All Atlantic tropical cyclones (including non-developing depressions) that existed between 1989 and 2000
- Only 24-h periods during which the system remained tropical and over water
- Predictors drawn from the NHC HURDAT best-track file and the operational SHIPS database

Analysis steps

1. Established the 30-kt / 24-h definition of RI from the intensity-change distribution
2. Compared the mean initial conditions of all RI periods with those of all non-RI periods and tested for statistical significance
3. Ranked the predictors by the magnitude of the difference between RI and non-RI samples
4. Constructed a simple probability estimator that counts how many of the five strongest predictors exceed chosen thresholds
5. Evaluated the resulting probability table and tested the method in real time during the 2001 Atlantic season under the Joint Hurricane Testbed

Results

- 31 % of all tropical cyclones, 60 % of all hurricanes, 83 % of all major hurricanes, and 100 % of category 4 and 5 hurricanes underwent at least one

RI episode during their lifetimes

- Relative to non-RI cases, RI cases on average:
  - Formed farther south and west and had a more westward component of motion
  - Were intensifying more rapidly in the previous 12 h
  - Were farther from their maximum potential intensity
  - Existed over warmer sea surface temperatures (SSTs) and in higher lower-tropospheric relative humidity

    💡 Lower-tropospheric relative humidity refers to the amount of moisture relative to saturation in the lowest part of the atmosphere :
    Temperature Dependence: Warm air holds more water vapour than cold air, meaning relative humidity changes inversely with temperature if moisture stays constant → Humidity decreases as temp increases
    Imagine a sponge:

    - Heating the air is like **stretching the sponge** to a larger size
    - If the amount of water inside the sponge stays the same, the stretched sponge is now **less saturated** (lower relative humidity) because it has the room to hold so much more

    It is **typically higher (often exceeding 70%) over oceans and moist land regions, and lower (under 50%) over major deserts**

  - Experienced weaker vertical shear and weaker upper-level trough forcing
  - Moved with the flow in a deeper atmospheric layer
- When zero of the five key thresholds were satisfied the estimated probability of RI was only 1 %; when all five were satisfied the probability rose to 41 %
- The simple technique was run in real time for the first time during the 2001 season

Conclusions

The authors conclude that large-scale environmental differences between RI and non-RI cases are statistically robust and can be turned into a transparent, operationally usable probability tool. They emphasise that the conditions favouring RI are broadly consistent with earlier theoretical and case-study results (low shear, high ocean heat content / warm SST, previous intensification, distance from MPI). The work established both the modern statistical definition of Atlantic RI and the conceptual foundation for all subsequent Rapid Intensification Index products.

Limitations

- Relies exclusively on large-scale SHIPS predictors; inner-core structure and high-resolution oceanic features are not directly included
- The simple threshold method does not account for interactions among predictors or for continuous (as opposed to binary) relationships
- Sample ends in 2000; later seasons and improved observational datasets are not included
- Only the Atlantic basin is examined
- Real-time performance is reported only anecdotally for the single 2001 season

Open Questions Left by the Paper

- Can more sophisticated statistical (or later machine-learning) methods extract additional skill from the same predictors?
- How stable are the identified thresholds and relative importance rankings when the sample is extended or when real-time analyses replace the SHIPS developmental fields?
- What is the relative contribution of oceanic versus atmospheric predictors once higher-resolution ocean heat-content data become routinely available?
- Can the same framework be successfully transferred to other basins?
- How should the still-modest absolute probabilities (maximum 41 %) be communicated to forecasters and emergency managers?

Useful Quotes

- “In this study, rapid intensification (RI) is defined as approximately the 95th percentile of over-water 24-h intensity changes of Atlantic basin tropical cyclones that developed from 1989 to 2000. This equates to a maximum sustained surface wind speed increase of 15.4 m s^{-1} (30 kt) over a 24-h period.” (Abstract)
- “It is shown that 31 % of all tropical cyclones, 60 % of all hurricanes, 83 % of all major hurricanes, and all category 4 and 5 hurricanes underwent RI at least once during their lifetimes.” (Abstract)
- “The probability of RI increases from 1 % to 41 % when the total number of thresholds satisfied increases from zero to five.” (Abstract)
- “The unexpected RI of these hurricanes so close to the coast underscores the need for improving our understanding of TC intensification, especially the RI process.” (Introduction)

💡 **How this paper supports the project**

- It supplies the exact definition of RI (30 kt in 24 h) that we adopt in Objective O2
- It lists the classic predictors (previous 12-h intensity change, distance from maximum potential intensity, vertical shear, SST, humidity) that we will extract from HURDAT2 and later from SHIPS
- Its simple threshold method is a transparent baseline we can re-implement and beat with XGBoost
- Its finding that almost every major hurricane experiences RI justifies focusing on this rare but high-impact event for catastrophe-risk applications

DeMaria & Kaplan (1994)

A Statistical Hurricane Intensity Prediction Scheme (SHIPS) for the Atlantic Basin

Mark DeMaria and John Kaplan. 1994. Weather and Forecasting, 9, 209–220.

Foundational statistical intensity model that became the operational baseline for Atlantic tropical-cyclone intensity guidance; still the reference against which later statistical, dynamical, and machine-learning intensity schemes are measured.

Overview

SHIPS is a multiple-linear-regression model that predicts Atlantic tropical-cyclone intensity change out to 72 h by combining 2 different kinds of information:

- Climatology and Persistence Predictors: What storms typically do at that time of year and how the storm has been changing recently
- A small set of Environmental Predictors: These are chosen for physical reasons, the most important of which are:
    - **Distance from maximum potential intensity:** How much stronger the storm could still become given the current sea-surface temperature. Large values mean the storm has plenty of room to intensify.
    - **Vertical wind shear:** The difference in wind speed and direction between the lower and upper atmosphere. Strong shear tilts and weakens the storm; weak shear allows it to strengthen.
    - **Upper-level eddy angular-momentum flux convergence:** Whether the high-altitude winds around the storm are adding spin (helps intensification) or removing spin (hinders it).

In jackknife tests that mimic real-time conditions, it reduces average intensity errors by 10–15 % relative to the purely climatological–persistence model SHIFOR, with the largest and statistically significant gains at 24–48 h.

Problem & Motivation

Operational Atlantic intensity forecasts in the early 1990s were “sorely lacking.” Track skill had improved steadily thanks to a hierarchy of statistical and dynamical models, but the only operational intensity guidance was SHIFOR, which used only climatology and persistence. Earlier attempts to add synoptic predictors produced only marginal gains. The authors therefore restricted the sample to oceanic cases, used high-resolution spline analyses, and selected predictors based on physical reasoning rather than pure statistical search.

Key Concepts & Terminology

- **SHIFOR**

  The operational climatology-and-persistence intensity model that served as the null baseline

  A weather forecast that only knows the average behaviour of storms on that calendar day and how the storm has been changing in the last 12 h; it has no information about the current atmospheric or oceanic environment

- **Maximum possible intensity (MPI)**

  An empirical upper bound on intensity derived from sea-surface temperature (SST) via the formula:

  $$MPI = 66.5 + 108.5e^{0.1813(\mathrm{SST}-30)}\ (\mathrm{kt})$$

  The key predictor Potential (POT) is MPI minus current intensity, averaged along the future track

  The temperature of the water sets a thermodynamic “ceiling”; a storm far below that ceiling has more room to intensify

- **Vertical shear (SHR)**

  Magnitude of the vector difference between the 850-mb and 200-mb winds, area-averaged in a 600-km radius and then averaged along the storm track out to 36 h

  A strong cross-wind that tilts and tears the storm apart.

- **Eddy angular-momentum flux convergence (REFC / PEFC)**

  Measures whether the large-scale flow at 200 mb is spinning up (positive) or spinning down (negative) the storm’s upper-level circulation

  Like a torque applied by the surrounding atmosphere, a positive torque helps the storm spin faster.

- **Jackknife verification**

  Leave-one-storm-out cross-validation: all forecasts for a given cyclone are removed, the regression is re-fit, and the withheld storm is predicted with the new coefficients. This approximates independent, real-time performance.

Methodology

Dataset construction

- 510 analyses of 49 Atlantic named tropical cyclones (primarily 1989–1992)
- Only times when the storm was over the ocean
- Intensity and position from the NHC best track
- Synoptic fields from cubic B-spline analyses
- SST from Levitus monthly climatology

Main regression procedure

- Separate multiple linear regressions at each lead time (12, 24, 36, 48, 60, 72 h)
- Predictors first chosen on physical grounds, then retained only if significant at the 95 % level for at least one lead time
- Final model retains 10 linear predictors + one quadratic term (POT^2)

Evaluation protocol

- Dependent verification, independent jackknife verification, and real-time simulation using operational intensities and VICBAR track forecasts
- Comparison baselines: SHIFOR and “no-change” forecast

Results

- Climatology + persistence alone explain 30–41 % of the variance of intensity change
- Adding the synoptic predictors raises explained variance to 36–54 %
- In jackknife tests that use operational intensities and VICBAR tracks, SHIPS average errors are 10–15 % smaller than SHIFOR; differences at 24, 36 and 48 h are significant at the 99 % level

- Maximum skill relative to SHIFOR occurs at 48 h
- For the top 10 % most rapidly intensifying cases SHIPS still under-forecasts but reduces the error by 33 % relative to SHIFOR

Conclusions

A carefully chosen set of synoptic predictors yields a statistically significant improvement over pure climatology and persistence. The three dominant predictors- intensity relative to the SST-determined MPI, vertical shear, and 12-h persistence are both physically sensible and quantitatively important at all lead times.

Limitations

- Sample size is modest and not fully homogeneous
- Only linear (plus one quadratic) relationships are admitted
- Shear and momentum-flux predictors are evaluated from the initial analysis only
- SST is climatological, not analysed in real time
- Intensity is treated as continuous even though best-track values are quantised to 5 kt
- Performance is weaker for baroclinic storms and for the most extreme rapid-intensification cases

Open Questions Left by the Paper

- Can non-linear or machine-learning methods extract additional skill from the same predictors?
- How much further improvement is possible once real-time SST and future shear evolution are supplied?
- Is a single set of regression coefficients optimal, or should separate models be trained for different intensity regimes?
- How should the inherent uncertainty of the intensity estimate itself be propagated into the forecast?

Useful Quotes

- "According to Sheets (1990), the skill of intensity forecasts of Atlantic tropical cyclones is 'sorely lacking.'" (Introduction)
- "The four primary predictors are 1) the difference between the current storm intensity and an estimate of the maximum possible intensity determined from the sea surface temperature, 2) the vertical shear of the horizontal wind, 3) persistence, and 4) the flux convergence of eddy angular momentum evaluated at 200 mb." (Abstract)
- "Results show that the average intensity errors are 10 %–15 % smaller than the errors from a model that uses only climatology and persistence (SHIFOR)…" (Abstract)

> 💡 **How this paper supports the project**
>
> - It is the origin of the SHIPS predictors that we plan to add in the second iteration (Objective O1)
> - The variables POT (distance from maximum potential intensity) and vertical shear remain among the strongest predictors of RI
> - Its jackknife (leave-one-storm-out) philosophy is the intellectual ancestor of the temporal / by-storm split we require in Objective O3
> - Its honest reporting of remaining errors on rapid-intensification cases sets the tone of scientific humility

Kaplan et al. (2010)

A Revised Tropical Cyclone Rapid Intensification Index for the Atlantic and Eastern North Pacific Basins

John Kaplan, Mark DeMaria, and John A. Knaff. 2010. Weather and Forecasting, 25, 220–241.

Foundational operational revision of the Rapid Intensification Index (RII) that became the National Hurricane Centre’s primary statistical guidance for RI probability forecasts in both the Atlantic and eastern North Pacific; still the baseline against which later statistical and machine-learning RI schemes are compared.

Overview

The authors revised the earlier Kaplan & DeMaria (2003) Rapid Intensification Index by applying linear discriminant analysis to a larger set of SHIPS predictors, expanding the product to three intensity-change thresholds (25, 30, and 35 kt in 24 h), and developing separate versions for the Atlantic and eastern North Pacific basins. The revised RII became operational at NHC before the 2008 season. Independent verification for 2006–2007 showed the probabilistic forecasts were generally skillful versus climatology and, when used deterministically, produced higher probability of detection than other operational intensity guidance, although false-alarm ratios remained high (especially in the Atlantic).

Problem & Motivation

Operational intensity forecasting, especially predicting rapid intensification, remained a major challenge in the late 2000s. The original five-predictor RII developed by Kaplan & DeMaria (2003) had been used operationally in the Atlantic since 2003, but it was limited to a single 30-kt threshold, used relatively simple statistical methods, and had no counterpart for the eastern North Pacific. The National Hurricane Centre had declared RI its top forecast priority. The present work therefore aimed to produce a more sophisticated, multi-threshold, dual-basin index that could be implemented operationally.

Key Concepts & Terminology

- **Rapid Intensification (RI)**

  Defined here as a 24-h increase in maximum sustained wind of at least 25, 30, or 35 kt. These thresholds correspond to roughly the 90th, 94th, and 97th percentiles of over-water intensity change in the Atlantic

  The difference between a storm that strengthens gradually and one that “jumps” one or two categories in a single day

- **Rapid Intensification Index (RII)**

  A statistical tool that converts a set of environmental and storm predictors into a probability that RI will occur in the next 24 h

- **Linear Discriminant Analysis (LDA)**

  The classification technique used to weight the predictors so that the linear combination best separates RI cases from non-RI cases

- **SHIPS predictors**

  The large-scale environmental variables routinely computed by the Statistical Hurricane Intensity Prediction Scheme

- **Probability of Detection (POD) / False Alarm Ratio (FAR)**
  - POD = fraction of actual RI events that were correctly forecast
  - FAR = fraction of forecast RI events that did not occur

Methodology

Data and climatology

- Atlantic and eastern North Pacific tropical and subtropical cyclones, 1989–2006 (over-water cases only)
- Updated RI climatology showing that the original 30-kt threshold is approximately the 94th percentile in the Atlantic
- Separate development samples for each basin and each of the three RI thresholds

Model development

- Predictors drawn from the operational SHIPS suite
- Linear discriminant analysis used to derive the optimal linear combination of predictors for each basin and each RI threshold
- Relative importance of predictors was allowed to differ between basins

Verification

- Independent forecasts for the 2006 and 2007 seasons

- Probabilistic skill assessed against climatology
- Deterministic skill assessed via POD and FAR and compared with all other operational intensity guidance available at the time

Results

- The revised RII became operational at the National Hurricane Center prior to the 2008 hurricane season
- Probabilistic forecasts were generally skillful relative to climatology in both basins
- When converted to deterministic forecasts, the RII produced higher POD than other operational intensity guidance:
  - Atlantic: POD 15–59 %, FAR 71–85 %
  - Eastern North Pacific: POD 53–73 %, FAR 53–79 %

  (ranges cover the three RI thresholds)
- The relative weighting of predictors differed markedly between the two basins
- Despite the improvements, both POD and FAR values remain modest, illustrating the intrinsic difficulty of RI prediction, especially in the Atlantic

Conclusions & Claims

The revised multi-threshold, dual-basin RII represents a clear advance over the original 2003 index and is skillful enough for operational use. The importance of individual predictors is basin-dependent and, even with the improved index, RI remains a low-POD / high-FAR forecast problem.

Limitations

- Relies on large-scale SHIPS predictors; inner-core processes and high-resolution oceanic features are only indirectly represented
- Independent verification is limited to two seasons (2006–2007)
- False-alarm ratios remain high, particularly in the Atlantic

- Linear discriminant analysis assumes linear separability; non-linear interactions are not captured
- Performance is weaker for the rarer, higher-threshold (35 kt) events

Open Questions Left by the Paper

- Can non-linear or machine-learning methods extract additional skill from the same (or expanded) predictor sets?
- How much further improvement is possible once real-time ocean heat content, microwave imagery, or higher-resolution model fields are incorporated?
- Are the basin differences in predictor importance stable over longer periods?
- What is the optimal way to communicate the still-high false-alarm rates to forecasters and emergency managers?

Useful Quotes

- “A revised rapid intensity index (RII) is developed for the Atlantic and eastern North Pacific basins… The revised RII became operational at the NHC prior to the 2008 hurricane season.” (Abstract)
- “The relative importance of the individual RI predictors is shown to differ between the two basins.” (Abstract)
- “Nevertheless, the modest POD and relatively high FAR of the RII and other intensity guidance demonstrate the difficulty of predicting RI, particularly in the Atlantic basin.” (Abstract)

> 💡 **How this paper supports the project**
>
> - This is the official operational baseline (SHIPS-RII) that we will benchmark against
> - It confirms that the Atlantic basin is more challenging than the eastern North Pacific (higher FAR, lower POD), so focusing on the Atlantic is the more valuable setting
> - Its use of multiple RI thresholds (25/30/35 kt) gives a ready-made sensitivity test we can run after the primary 30-kt model is finished

- The documented high false-alarm rates remind that any claim of “skill” must be accompanied by clear reporting of precision and recall at the chosen operating point

Mercer & Grimes (2017)

Atlantic Tropical Cyclone Rapid Intensification Probabilistic Forecasts from an Ensemble of Machine Learning Methods

Andrew Mercer and Alexandria Grimes. 2017. Procedia Computer Science, 114, 333–340.

Early demonstration that an ensemble of classical machine-learning classifiers can produce probabilistic RI forecasts for Atlantic tropical cyclones that match or exceed the skill of the then-operational linear discriminant analysis (LDA) scheme, while revealing a higher theoretical skill ceiling.

Overview

The authors built one of the first machine-learning ensembles (support vector machines, artificial neural networks, and random forests) to issue probabilistic forecasts of rapid intensification of Atlantic tropical cyclones. After careful feature selection and configuration tuning, the ensemble’s probabilistic skill matched the operational LDA-based Rapid Intensification Index, yet the best individual members and the ensemble upper bound exceeded climatology by 30%, roughly double the skill of the existing operational scheme at the time.

Problem & Motivation

Rapid intensification remained one of the most difficult aspects of tropical-cyclone forecasting. At the time of the study, the best operational guidance (the SHIPS Rapid Intensification Index based on linear discriminant analysis) was only about 15 % better than simple climatology. No prior work had systematically tested whether modern machine-learning methods could improve probabilistic RI forecasts.

Key Concepts & Terminology

- **Rapid Intensification (RI)**

  An increase of at least 30 knots in the 1-minute maximum sustained wind within a 24-hour period

  A hurricane that “jumps” from Category 1 to Category 3 in a single day

- **Linear Discriminant Analysis (LDA)**

  The statistical method used in the operational SHIPS Rapid Intensification Index at the time

- **Proxy forecast model data**

  Fields that approximate what an operational numerical weather prediction model would have produced

- **Backwards elimination feature selection**

  Start with a large pool of candidate predictors and iteratively remove the least useful ones until performance stops improving

- **Bootstrap-based cross-validation**

  Repeatedly resample the training data with replacement to evaluate how stable and generalisable each model configuration is

- **Probabilistic ensemble weighting**

  Each machine-learning member produces its own RI probability; these probabilities are combined with weights proportional to each member’s cross-validated skill

Methodology

Dataset

- Atlantic tropical cyclones, 1985–2011
- Only times when the storm was over water
- Predictors came from proxy numerical-model forecast fields, the predictors already used by the operational LDA/SHIPS-RI scheme, and observed storm track and intensity information

Three-step optimisation

1. Feature selection: Backwards elimination on the combined predictor pool
2. Model configuration search: We trained and ranked many variants of three algorithms (support vector machines, artificial neural networks, and random forests) using bootstrap cross-validation
3. Ensemble construction: Selected configurations produced individual probabilistic forecasts that were combined into a single ensemble probability weighted by each member’s cross-validation skill

Results

- The final ensemble’s average probabilistic skill was essentially equal to that of the operational LDA scheme
- The upper end of the ensemble’s performance distribution exceeded 30 % improvement over climatology, approximately twice the skill of the then-current operational system
- Individual well-tuned members of the support-vector-machine, neural-network, and random-forest families all contributed useful skill

Conclusions

Machine-learning methods can produce useful probabilistic RI forecasts for Atlantic tropical cyclones. While the ensemble's mean skill matched the operational LDA baseline, configurations that more than doubled skill relative to climatology suggested further gains were possible.

Limitations

- Relied on proxy rather than true real-time forecast fields
- Training period ends in 2011
- Feature selection and model tuning were performed on the same overall dataset
- Only Atlantic basin cases were considered

Open Questions Left by the Paper

- How much of the apparent skill ceiling (>30 % over climatology) can be realised in true real-time operational forecasts?
- Would the same ensemble architecture transfer successfully to other basins?
- Can newer deep-learning architectures or additional satellite-based predictors push skill still higher?

Useful Quotes

- "Atlantic tropical cyclone (TC) rapid intensification (RI) continues to be a major forecasting challenge, with forecast skill scores only about 15% better than climatology." (Abstract)
- "Resulting probabilistic forecasts were in line with the current LDA method, though the upper skill limit of the ensemble exceeded 30% improvement over climatology, which far exceeds the current LDA scheme." (Abstract)

💡 **How this paper supports the project**

- It is the first published demonstration that classical machine-learning methods can match or beat the operational SHIPS-RII on Atlantic RI. XGBoost model is a natural modern continuation of this line of work
- Their use of an ensemble and careful feature selection provides a useful point of comparison for the single-model, SHAP-interpretable approach
- The reported upper skill ceiling of >30 % over climatology gives a realistic target range; anything higher would be extraordinary and should be treated with caution

An et al. (2026)

Machine Learning Based Prediction of Tropical Cyclone Rapid Intensification in the Western North Pacific: Importance of Data Augmentation and Loss Function

Sanghyeok An, Jiwon Jeong, Seok-Woo Son, Hyungjun Kim, Jee-Hoon Jeong, and Jin-Ho Yoon. 2026. Journal of Geophysical Research: Machine Learning and Computation, 3, e2025JH000876.

A modern tabular deep-learning approach to the classic rapid-intensification (RI) prediction problem demonstrates practical solutions to severe class imbalance and provides interpretable feature rankings that connect back to physical understanding.

Overview

The authors build a TabNet classifier to predict whether a Western North Pacific tropical cyclone will undergo rapid intensification (RI: ≥30 kt increase in 24 h). Because RI cases are rare (~1:4 ratio), they systematically test Synthetic Minority Oversampling Technique (SMOTE) at different ratios and three loss functions. The combination of 2:1 SMOTE + balanced cross-entropy yields the best Critical Success Index. TabNet's built-in attention mechanism ranks sea-surface temperature as the most important predictor, followed by 850-hPa temperature and specific humidity. Performance varies strongly from year to year and is weakest in years with very few RI events (e.g. 2020).

Problem & Motivation

Rapid intensification is one of the hardest parts of tropical-cyclone forecasting. A major practical obstacle is class imbalance: RI is a rare event. In the 1977–2021 Western North Pacific record the ratio of non-RI to RI cases is roughly 4.2 : 1. Most machine-learning models trained on such data simply learn to predict "no RI" almost all the time. The paper's central contribution is showing how careful data augmentation (SMOTE) and loss-function design can overcome this imbalance while keeping the model interpretable.

Key Concepts & Terminology

- **Rapid Intensification (RI)**

  An increase of at least 30 knots (~15.4 m/s) in maximum sustained wind speed within 24 hours

- **Class imbalance**

When one class (non-RI) vastly outnumbers the other (RI). Models tend to ignore the rare class

- **SMOTE (Synthetic Minority Oversampling Technique)**

  Creates new artificial RI examples by interpolating between existing RI cases and their nearest neighbours in feature space

  Instead of simply photocopying the few rare photos you have, you create realistic new photos by blending features of the existing ones

- **TabNet**

  A deep-learning architecture designed for tabular data. It uses sequential decision steps with sparse attention masks, so it both selects important features at each step and remains interpretable

- **Balanced cross-entropy / Focal loss**

  Modified loss functions that give higher weight to the rare RI class (or to hard-to-classify examples)

- **Critical Success Index (CSI)**

  A single number that penalises both missed RI events and false alarms

$$\frac{\text{True Positives}}{\text{True Positives} + \text{False Positives} + \text{False Negatives}}$$

Methodology

Dataset construction

- Best-track data from the Japan Meteorological Agency (1977–2021) for Western North Pacific tropical cyclones
- Environmental predictors extracted from ERA5 reanalysis
- Chronological split: training 1977–2007, validation 2008–2014, test 2015–2021

Model

TabNet classifier

Handling imbalance

- SMOTE applied only to the training set at four ratios: 4:1, 3:1, 2:1, 1:1

- Three loss functions tested: ordinary cross-entropy, balanced cross-entropy, and focal loss
- The 2:1 SMOTE + balanced cross-entropy combination produced the highest validation CSI

Evaluation

CSI (primary), F1-score, Pierce Skill Score, Probability of Detection, False Alarm Ratio. Validation and test sets kept their natural (imbalanced) class ratios

Results

- The combination of 2:1 SMOTE and balanced cross-entropy gave the highest CSI
- TabNet's attention mechanism ranked sea-surface temperature as the single most important feature, followed by temperature and specific humidity at 850 hPa
- Performance showed clear year-to-year variation; 2020 was particularly poor, mainly because of the very small number of RI cases that year

Conclusions

TabNet, when paired with appropriate data augmentation and a carefully chosen loss function, can significantly improve RI prediction skill for Western North Pacific tropical cyclones while remaining interpretable. Sample size (especially the absolute number of RI cases) is a critical limiting factor

Limitations

- Results are basin-specific (Western North Pacific only)
- Performance remains sensitive to the absolute number of RI cases in a given year
- SMOTE creates synthetic samples whose physical realism is not guaranteed
- Only environmental predictors from ERA5 are used; inner-core structure or satellite imagery is not included

Open Questions Left by the Paper

- How transferable is the 2:1 SMOTE + balanced-cross-entropy recipe to other basins?
- Can the same TabNet architecture usefully incorporate high-resolution satellite or radar fields?
- What is the optimal way to update the model as new seasons add more RI cases?

Useful Quotes

- "The most significant challenge in predicting RI is the severe class imbalance between rapid and non-rapid intensification cases, typically 4.2:1 ratio based on 1977–2021 records." (Abstract)
- "The combination of balanced cross entropy and 2:1 SMOTE achieved the highest performance." (Abstract)
- "Leveraging the TabNet's interpretability, sea surface temperature was identified as the most important factor, followed by temperature and specific humidity at 850 hPa." (Abstract)

> 💡 **How this paper supports the project**
>
> - It is the most recent and most explicit demonstration that class-imbalance handling is the true technical crux of RI modelling
> - Their systematic comparison of SMOTE ratios and loss functions gives a ready-made experimental template we can adapt for XGBoost (scale_pos_weight, threshold tuning, optional SMOTE
> - Their finding that performance collapses in years with few RI cases warns to examine year-by-year skill on the held-out seasons and to report it honestly
> - The ranking of sea-surface temperature as the top predictor reinforces the physical variables we already plan to include from SHIPS

Chen et al. (2020)

Machine Learning in Tropical Cyclone Forecast Modeling: A Review

Rui Chen, Weimin Zhang, and Xiang Wang. 2020. Atmosphere, 11, 676.

Comprehensive survey of pure data-driven and hybrid ML approaches for tropical-cyclone genesis, track, intensity, associated hazards, and numerical-model improvement; useful baseline for any project that applies modern ML to extreme-weather prediction.

Overview

Traditional dynamical and statistical TC forecast models still struggle with genesis, intensity (especially rapid intensification), and associated hazards because the underlying physics are incompletely understood and the relationships are highly non-linear. Machine learning, ranging from classical methods (decision trees, SVMs, random forests) to deep networks (CNNs, RNNs, ConvLSTMs, GANs), has already produced measurable gains across all major forecast tasks by learning directly from multi-source data. The paper frames these advances as both an opportunity (vast unused data + still-under-exploited algorithms) and a challenge (short lead times, limited interpretability, scarce in-situ observations)

Problem & Motivation

Tropical cyclones remain one of the most destructive weather systems. After a century of research, track forecasts have improved substantially, but genesis, intensity change (especially rapid intensification), wind fields, rainfall, and storm-surge forecasts lag. Machine learning is positioned as a complementary route that can either learn pure data-driven mappings when the physics are poorly known or improve existing numerical models via better parameterisation, data-quality control, or post-processing.

Key Concepts & Terminology

The review covers the full spectrum of ML applied to Tropical Cyclones; the most relevant for the project are the intensity and RI sections, which reiterate the same predictors and challenges already covered above

Key takeaway for the project is that, as of 2020, pure data-driven and hybrid ML approaches had already shown promise on RI, but interpretability, lead-time, and sample-size limitations remained open problems which are gaps the concept note aims to address with an honest, SHAP-interpretable, temporally validated XGBoost model

💡 **How this paper supports the project**

- It situates the work inside the broader literature and confirms that intensity/RI remains the hardest sub-problem
- It catalogues the classical ML methods (decision trees, random forests, SVMs) whose strengths and weaknesses we can contrast with the chosen gradient-boosted trees
- Its emphasis on the still-under-exploited potential of multi-source data and the continuing need for interpretable models directly legitimises the design choices in the concept note (HURDAT2 + SHIPS, SHAP, honest limitations)

How the Literature Maps onto the Project Objectives

The Objective	Papers that directly inform it	Concrete action we can take
O1. Assemble clean training data from HURDAT2 (+ SHIPS)	Kaplan & DeMaria 2003; DeMaria & Kaplan 1994; Kaplan et al. 2010	Use the exact 30-kt / 24-h definition; extract the classic predictors (POT, shear, previous 12-h change, SST, humidity) that every successful model has retained.
O2. Define the RI label rigorously and quantify class imbalance	All papers	Report the exact ratio in the Atlantic sample; treat the imbalance as the central modelling challenge (An et al. 2026).
O3. Train a gradient-boosted classifier with	An et al. 2026; Mercer & Grimes	Adopt 2:1-style oversampling or scale_pos_weight; never use a random

The Objective	Papers that directly inform it	Concrete action we can take
explicit imbalance handling and temporal validation	2017; Kaplan et al. 2010	row split; evaluate with PR-AUC / CSI / POD–FAR exactly as the literature does.
O4. Interpret the model with SHAP	An et al. 2026 (TabNet attention); Chen et al. 2020 (call for interpretability)	Expect SST / ocean heat content and previous intensity change to rank high; any surprise ranking is itself a publishable observation.
O5. Deploy as a small Flask API	(Engineering choice; literature supplies the scientific payload)	The model deployed will be the first publicly available, fully documented, SHAP-explained RI probability service based on open data.
O6 (stretch). Test early-warning signals (critical slowing down)	Concept note itself + Scheffer et al. 2009	The literature has only begun to ask whether RI and non-RI storms occupy distinct dynamical regimes; therefore, the exploratory variance/autocorrelation analysis is genuinely novel.